

HERA-PdM: Lessons from Deploying Uncertainty-Gated Edge AI for Predictive Maintenance on Two Bearing Datasets

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Abstract—Deploying AI for predictive maintenance (PdM) requires deciding where inference runs, how far predictions can be trusted, and when humans must act. Hierarchical systems usually escalate from edge to cloud on anomaly thresholds. We study HERA-PdM, in which cross-validated conformalized quantile regression (CQR-CV+) gives remaining-useful-life (RUL) intervals, the edge escalates only when its interval does not fix a unique maintenance action, and high-impact actions are SHAP-supported recommendations for human approval. On 17 FEMTO and 15 XJTU-SY run-to-failure bearings (bearing-grouped cross-validation, three seeds, tuned baselines), CQR-CV+ raised coverage to 0.94–0.96 and 0.93–0.99 (nominal 0.90). Versus point estimates with fixed thresholds, calibrated decisions cut critical snapshots left without action from 49.5% to 0.6% (FEMTO) and 34.2% to 9.2% (XJTU-SY), at a false-maintenance cost. On gradually degrading XJTU-SY bearings, escalating 79.1% of snapshots matched always-cloud safety with less false maintenance (53.2% vs. 69.1%), a gain not robust across bearings; on abrupt FEMTO failures it offered no benefit. The decision-sufficiency gate performed no better than an interval-width gate, all calibrated policies raised maintenance alarms prematurely for most bearings, and conformal risk control met a per-bearing lateness target only by alarming early. Uncertainty-gated hierarchies thus pay off only when degradation is observable early.

Index Terms—AI deployment, predictive maintenance, edge AI, conformal prediction, explainable AI, human-in-the-loop, Digital Twin

I. INTRODUCTION

Predictive maintenance (PdM) estimates the remaining useful life (RUL) of machinery from condition data so that interventions are scheduled before failure [1]. Deploying it on a factory floor raises three questions that accuracy benchmarks ignore: *where* inference runs, since vibration is sensed at the machine and edge–cloud architectures keep routine inference local [2]; *how far* a prediction can be trusted, which calls for calibrated uncertainty rather than point estimates [3], [4]; and *who* acts on it, since Industry 5.0 keeps humans in charge of consequential decisions [5].

Hierarchical PdM systems that combine edge devices, Digital Twins (DTs) and agents already exist [6], [7]; they escalate from edge to cloud on anomaly thresholds [6] or uncalibrated confidence [7]. Calibrated conformal deferral with guarantees has been developed for classification cascades [8] but not for RUL-based maintenance. We therefore ask *when* conformally calibrated RUL intervals computed on the edge are a useful escalation signal, and what they cost in safety, false maintenance, alarm timing, communication and latency.

Key findings. On two public bearing datasets with contrasting degradation behaviour, (i) calibration removes most unsafe decisions of point-threshold maintenance; (ii) uncertainty-gated escalation pays off only when degradation is observable early; (iii) the precise gating rule matters less than the interval width; and (iv) snapshot-level guarantees do not yield timely bearing-level alarms, while bearing-level guarantees need larger fleets than public datasets provide.

Contributions:

- 1) **HERA-PdM**, a reproducible four-layer framework with a *decision-sufficiency* gate that escalates only when the edge RUL interval straddles a maintenance-decision boundary, bounding locally resolved unsafe decisions by the edge miscoverage level, and CQR-CV+ calibration on all non-test assets.
- 2) A two-dataset, bearing-grouped benchmark of ten tuned quantile models with and without DT context, showing that CQR-CV+ restores nominal coverage and that larger fog/cloud models help only under gradual degradation.
- 3) Snapshot-level, bearing-level and sensitivity evidence, with bootstrap intervals over bearings, that turns findings (i)–(iv) into six deployment lessons.

II. RELATED WORK AND RESEARCH GAP

RUL prediction. Deep sequence models such as temporal convolutional networks [9] are standard RUL estimators; reviews [1] treat health-stage division, i.e. detecting when degradation becomes observable, as a prerequisite for prognosis, a property our two datasets [10], [11] contrast sharply. *Hierarchical PdM.* Ren et al. [2] run a lightweight TCN at the edge and refine RUL in the cloud; HAMA [6] escalates from a z-score edge pre-filter to fog ensembles with SHAP and generated operator text; EC-Distill-ZeroDiag [7] offloads uncertain fault-diagnosis cases from an edge agent to a cloud model. *Calibrated uncertainty.* Conformalized quantile regression (CQR) [3] and CV+ [12] give distribution-free coverage; conformal RUL intervals exist for turbfans [4] and bearings [13], but without a decision or routing layer. Conformal risk control [14] bounds expected monotone losses, and conformal alignment certifies edge–cloud cascades for classification [8].

Gap. To the best of our audit of 42 papers (Table I), no study uses conformal RUL intervals computed on the edge as the escalation rule of a hierarchical PdM system, relates escalation to maintenance-decision boundaries, and evaluates

TABLE I
CLOSEST RELATED WORK (CODED FROM FULL TEXT, ABSTRACTS OR PUBLIC CODE). ✓: PRESENT, ○: PARTIAL, -: ABSENT.

Work	DT	Edge	Agents	RUL	Calib.	UQ	Routing	XAI	HIL
Ren et al. [2]	-	✓	-	✓	-	-	-	-	-
HAMA [6]	-	✓	✓	○	-	-	○ anomaly	✓	○
EC-Distill [7]	-	✓	✓	-	-	-	✓ conf.	○	-
CAb cascade [8]	-	✓	-	-	✓	-	✓	-	-
Wang et al. [13]	-	-	-	✓	✓	-	-	-	-
HERA-PdM	○	✓	✓	✓	✓	-	✓	✓	✓

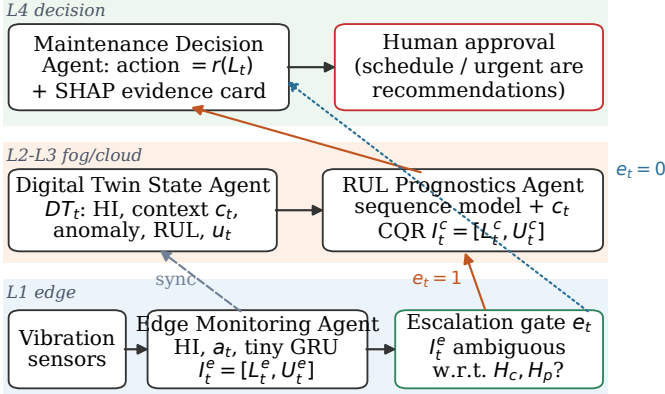


Fig. 1. HERA-PdM. The edge agent computes a calibrated RUL interval I_t^e ; the gate escalates only if I_t^e is not decision-sufficient; the fog/cloud agent uses the DT context c_t ; schedule/urgent actions need human approval.

it against the gates of existing systems on public run-to-failure data with different degradation regimes. The contribution is methodological and evaluative; the architecture is not new.

III. HERA-PdM METHODOLOGY

Fig. 1 shows the four layers. “Agents” are software components with fixed responsibilities and messages; no language model is used anywhere.

Edge Monitoring Agent. Each vibration snapshot is reduced to $x_t \in \mathbb{R}^{28}$ (per axis: RMS, kurtosis, skewness, peak-to-peak, crest, impulse and shape factors, spectral centroid, six log band energies), z-scored against the asset’s first snapshots and compressed by $\text{sign}(z) \log(1+|z|)$. The health index is $h_t = \log(1 + \frac{1}{d} \sum_j |z_{t,j}|)$; the anomaly score a_t is the smoothed h_t standardized by the asset’s raw baseline, with alarm threshold τ_a at the 99th percentile of healthy training snapshots. A tiny model maps a short window to three RUL quantiles.

Digital Twin State Agent. The twin is a synchronized state record $DT_t = \{\text{condition, sensor state, } h_t, \text{ context } c_t, \hat{R}_t, I_t, u_t, \text{ anomaly state}\}$ updated from a 12-byte sync (t, h_t, a_t) per snapshot; $c_t \in \mathbb{R}^6$ holds fast/slow averages of h_t , their difference, running maximum, slope and cumulative excess over baseline. Absolute operating time is excluded because lifetimes vary by an order of magnitude.

RUL Prognostics Agent. A sequence model g_ϕ maps a longer window X_t and c_t to non-crossing quantiles

$(\hat{q}_t^{\alpha/2}, \hat{q}_t^{0.5}, \hat{q}_t^{1-\alpha/2})$ trained with the pinball loss on $y_t = \min(\text{RUL}_t, R_{\max})$.

Calibration (CQR-CV+). The non-test assets are split into $K=4$ condition-balanced groups; inner model $g^{(-k)}$ is trained without group k , which yields out-of-fold scores $s_i = \max(\hat{q}_i^{\alpha/2} - y_i, y_i - \hat{q}_i^{1-\alpha/2})$ for every non-test snapshot i in group $k(i)$. Following CV+ [12] with CQR scores [3],

$$L_t = Q_\alpha^- \{\hat{q}_t^{\alpha/2, (-k(i))} - s_i\}, \quad (1)$$

$$U_t = Q_{1-\alpha}^+ \{\hat{q}_t^{1-\alpha/2, (-k(i))} + s_i\}, \quad (2)$$

where Q_α^- and $Q_{1-\alpha}^+$ are the $[\alpha(n+1)]$ -th and $[(1-\alpha)(n+1)]$ -th smallest values over the n calibration snapshots; $\hat{R}_t$ is the mean inner median. CV+ guarantees coverage $\geq 1 - 2\alpha$ (typically $\approx 1 - \alpha$) if assets are exchangeable; snapshots within an asset are dependent, so the guarantee is approximate and we test it. Split CQR on three held-out bearings is the baseline calibrator.

Decision and escalation. With critical and planning horizons H_c and H_p , the action of a value v is $r(v) = \text{URGENT}$ if $v \leq H_c$, SCHEDULE if $H_c < v \leq H_p$, else CONTINUE; the Maintenance Decision Agent acts on $\pi_t = r(L_t)$ (CONTINUE becomes INSPECT if $a_t > \tau_a$). SCHEDULE/URGENT are recommendations with an explanation card (RUL, interval, anomaly evidence, top-3 SHAP [15] contributors, rationale) that require approval. With $B = \{H_c, H_p\}$, the edge escalates if its interval is ambiguous or contradicts anomaly evidence:

$$e_t = \mathbb{I}[\exists h \in B: L_t^e \leq h < U_t^e] \vee \mathbb{I}[a_t > \tau_a \wedge r(L_t^e) = C], \quad (3)$$

where C denotes CONTINUE. A local CONTINUE needs $L_t^e > H_p \geq H_c$, so a critical asset ($y_t \leq H_c$) is left running locally only if $y_t \notin I_t^e$:

$$\Pr(e_t=0, \pi_t=\text{CONTINUE}, y_t \leq H_c) \leq \Pr(y_t \notin I_t^e), \quad (4)$$

and false local URGENT actions are bounded likewise. Unlike a width threshold, the gate ignores wide intervals inside one action region and escalates narrow ones crossing a boundary.

Cost model. Payloads are counted in float32: always-cloud streams x_t (112 B/snapshot); HERA sends the 12 B sync plus the untransmitted part of the window when $e_t=1$. Latency is edge time + $e_t(\text{RTT} + \text{cloud time})$ with an assumed RTT of 50 ms.

IV. EXPERIMENTAL SETUP

Data. FEMTO-ST/PRONOSTIA [10]: 17 bearings under three conditions, 0.1 s snapshots every 10 s (24,889 snapshots), full test trajectories; several bearings fail abruptly. XJTU-SY [11]: 15 bearings under 35 Hz/12 kN, 37.5 Hz/11 kN and 40 Hz/10 kN, 1.28 s snapshots every minute (9,216 snapshots), mostly gradual degradation. RUL caps are $R_{\max}=3000$ s (FEMTO) and 6000 s (XJTU-SY); $H_c=0.2R_{\max}$, $H_p=0.6R_{\max}$; windows are 16/64 and 8/32 snapshots (edge/cloud).

Protocol. Condition-balanced bearing-grouped folds (4 for FEMTO, 5 for XJTU-SY) test every bearing once. Each architecture’s hyper-parameters (width 32–128, 8 or 25 epochs,

TABLE II

RUL ACCURACY AND CALIBRATION ON HELD-OUT BEARINGS (MEAN $\pm$ STD OVER 3 SEEDS; NOMINAL COVERAGE 0.90). RAW: UNCALIBRATED QUANTILES; SPLIT: CQR CALIBRATED ON 3 HELD-OUT BEARINGS PER FOLD; CV+: CQR-CV+ CALIBRATED ON ALL NON-TEST BEARINGS. MARKS: ARCHITECTURE SELECTED ON INNER-VALIDATION PINBALL LOSS FOR FEMTO ($\dagger$) AND XJTU-SY (*).

Model	FEMTO/PRONOSTIA (17 bearings, $R_{\max}=3000$ s)					XJTU-SY (15 bearings, $R_{\max}=6000$ s)				
	RMSE (s)	MAE (s)	Raw cov.	Cov.	MPIW (s)	RMSE (s)	MAE (s)	Raw cov.	Cov.	MPIW (s)
Edge-CNN* (split)	1114 $\pm$ 40	851 $\pm$ 19	0.37	0.88 $\pm$ 0.02	2080	2278 $\pm$ 389	1859 $\pm$ 330	0.65	0.97 $\pm$ 0.00	3716
Edge-GRU $\dagger$ (split)	820 $\pm$ 34	518 $\pm$ 62	0.71	0.84 $\pm$ 0.04	1877	2531 $\pm$ 642	2004 $\pm$ 601	0.85	0.87 $\pm$ 0.08	4184
LSTM (split)	1208 $\pm$ 118	772 $\pm$ 102	0.50	0.85 $\pm$ 0.04	2263	1516 $\pm$ 127	886 $\pm$ 104	0.90	0.91 $\pm$ 0.01	2436
LSTM+DT $\dagger$ (split)	1124 $\pm$ 48	751 $\pm$ 54	0.37	0.84 $\pm$ 0.04	2154	1698 $\pm$ 58	1130 $\pm$ 68	0.89	0.88 $\pm$ 0.03	3107
GRU (split)	1171 $\pm$ 114	751 $\pm$ 84	0.40	0.86 $\pm$ 0.03	2151	1516 $\pm$ 39	1027 $\pm$ 73	0.89	0.89 $\pm$ 0.02	2404
GRU+DT* (split)	1140 $\pm$ 89	771 $\pm$ 66	0.32	0.89 $\pm$ 0.03	2286	1647 $\pm$ 94	1226 $\pm$ 153	0.86	0.83 $\pm$ 0.06	3042
TCN (split)	1159 $\pm$ 70	827 $\pm$ 72	0.16	0.85 $\pm$ 0.02	2241	1666 $\pm$ 119	1303 $\pm$ 145	0.52	0.95 $\pm$ 0.01	3691
TCN+DT (split)	1153 $\pm$ 136	855 $\pm$ 115	0.16	0.89 $\pm$ 0.01	2456	1496 $\pm$ 118	1100 $\pm$ 138	0.62	0.81 $\pm$ 0.11	3477
CNN-LSTM (split)	1106 $\pm$ 74	674 $\pm$ 34	0.40	0.85 $\pm$ 0.02	2119	1335 $\pm$ 3	772 $\pm$ 21	0.85	0.90 $\pm$ 0.01	2416
CNN-LSTM+DT (split)	1108 $\pm$ 39	747 $\pm$ 54	0.28	0.89 $\pm$ 0.01	2355	1446 $\pm$ 30	895 $\pm$ 33	0.84	0.91 $\pm$ 0.00	2686
LSTM+DT (MSE point)	1090 $\pm$ 161	780 $\pm$ 129	–	–	–	1656 $\pm$ 41	1134 $\pm$ 54	–	–	–
GRU+DT (MSE point)	1166 $\pm$ 121	858 $\pm$ 92	–	–	–	1689 $\pm$ 115	1293 $\pm$ 159	–	–	–
TCN+DT (MSE point)	1232 $\pm$ 110	925 $\pm$ 89	–	–	–	1539 $\pm$ 95	1156 $\pm$ 107	–	–	–
CNN-LSTM+DT (MSE point)	1135 $\pm$ 45	825 $\pm$ 37	–	–	–	1468 $\pm$ 29	897 $\pm$ 14	–	–	–
Selected edge (CV+)	753 $\pm$ 5	501 $\pm$ 22	0.75	0.94 $\pm$ 0.01	2193	1938 $\pm$ 251	1677 $\pm$ 232	0.69	0.99 $\pm$ 0.01	4134
Selected cloud+DT (CV+)	892 $\pm$ 27	665 $\pm$ 14	0.28	0.96 $\pm$ 0.02	2506	1422 $\pm$ 38	1077 $\pm$ 59	0.91	0.93 $\pm$ 0.01	3030
Selected cloud, no DT (CV+)	954 $\pm$ 16	653 $\pm$ 13	0.35	0.95 $\pm$ 0.02	2551	1305 $\pm$ 54	757 $\pm$ 66	0.90	0.94 $\pm$ 0.01	2337

TABLE III

HIERARCHICAL POLICIES, ABLATIONS AND SYSTEM COST (MEAN $\pm$ STD OVER 3 SEEDS, ALL HELD-OUT BEARINGS). ESC.: ESCALATED SNAPSHOTS; UNSAFE: CRITICAL SNAPSHOTS ($y \leq H_c$) LEFT WITHOUT A MAINTENANCE ACTION; FM: FALSE MAINTENANCE ON HEALTHY SNAPSHOTS ($y > H_p$); B/s: PAYLOAD BYTES PER SNAPSHOT (ANALYTIC). $\dagger$ ESCALATION BUDGET MATCHED TO HERA ON CALIBRATION DATA.

Policy	FEMTO/PRONOSTIA					XJTU-SY				
	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s
Always-edge	0.0	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	752	0	0.0	0.0 $\pm$ 0.0	77.1 $\pm$ 7.1	1485	0
Always-cloud	100.0	0.0 $\pm$ 0.0	96.9 $\pm$ 2.7	891	112	100.0	9.2 $\pm$ 1.7	69.1 $\pm$ 6.1	1421	112
Cloud point + fixed thr.	100.0	49.5 $\pm$ 2.7	21.7 $\pm$ 6.8	1090	112	100.0	34.2 $\pm$ 9.2	8.3 $\pm$ 2.1	1689	112
HERA-full	97.1	0.6 $\pm$ 1.1	95.0 $\pm$ 0.5	892	123	79.1	9.2 $\pm$ 1.7	53.2 $\pm$ 13.3	1381	104
HERA w/o anomaly term	97.1	0.6 $\pm$ 1.1	95.0 $\pm$ 0.5	892	123	79.1	9.2 $\pm$ 1.7	53.2 $\pm$ 13.3	1381	104
HERA-no-DT	97.1	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	953	123	79.1	12.1 $\pm$ 4.3	25.5 $\pm$ 7.7	1303	104
HERA-no-UQ (point margin)	48.1	56.2 $\pm$ 5.3	15.1 $\pm$ 2.8	989	89	67.8	34.2 $\pm$ 9.2	8.0 $\pm$ 2.2	2062	91
HERA-fixed (anomaly trig.)	3.9	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	760	27	5.4	0.0 $\pm$ 0.0	77.1 $\pm$ 7.1	1541	20
Width gate $\dagger$	97.2	0.6 $\pm$ 1.1	95.1 $\pm$ 0.5	892	123	79.2	9.2 $\pm$ 1.7	53.3 $\pm$ 13.3	1381	104
Linear score gate $\dagger$	94.3	0.6 $\pm$ 1.1	95.7 $\pm$ 0.9	892	121	77.9	9.2 $\pm$ 1.7	53.8 $\pm$ 13.3	1378	103
Random gate $\dagger$	96.4	0.0 $\pm$ 0.0	96.8 $\pm$ 2.6	887	124	71.6	6.5 $\pm$ 1.4	71.4 $\pm$ 6.3	1441	124

input noise) and the edge/cloud architecture choice were selected on inner validation bearings from training bearings only (FEMTO: Edge-GRU/LSTM+DT; XJTU-SY: Edge-CNN/GRU+DT). Three seeds per model; metrics are pooled over test bearings and reported as mean $\pm$ std; paired Wilcoxon tests use bearings as units.

Baselines and metrics. LSTM, GRU, TCN [9] and CNN-LSTM with and without DT context, MSE point models; always-edge, always-cloud and point-with-fixed-threshold policies; ablations without uncertainty (point margin), DT or anomaly term; anomaly-trigger (as in [6]), width (confidence-style deferral), linear-score and random gates at HERA’s budget. We report RMSE, MAE, coverage (Cov.), mean interval width (MPIW), *unsafe rate* (critical snapshots with CONTINUE/INSPECT), false maintenance (FM, SCHEDULE/URGENT when $y_t > H_p$), escalation, payload, latency

(single-thread host CPU, Apple M2 Pro; no energy measured) and, per bearing, the timing of the first sustained maintenance alarm.

V. RESULTS AND DISCUSSION

RQ1: accuracy. Table II shows opposite regimes. On FEMTO, the 2.5k-parameter edge model was most accurate (RMSE 753 s with CQR-CV+ vs. 892 s for LSTM+DT), because abrupt failures leave larger models nothing to exploit. On XJTU-SY, fog/cloud models were clearly better (best split-trained 1335 s, CNN-LSTM) than the selected edge model (2278 s). DT context lowered RMSE for 3 of 4 architectures on FEMTO but only 1 on XJTU-SY; cross-fitted ensembles (CV+) also improved point accuracy.

RQ2: calibration. Split CQR with three calibration bearings under-covered on FEMTO (0.84–0.89). CQR-CV+, which cal-

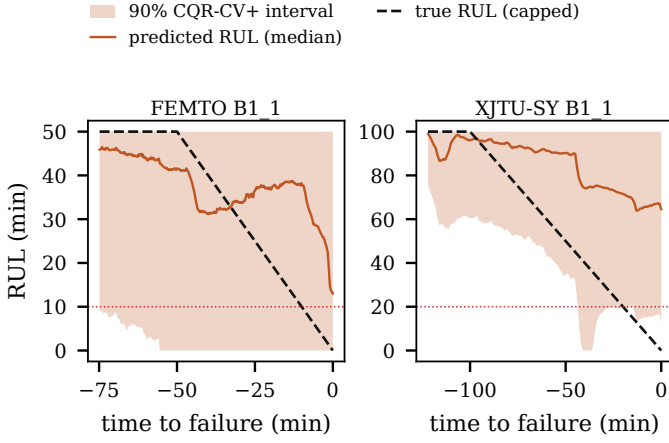


Fig. 2. Fog/cloud median RUL and 90% CQR-CV+ interval before failure (seed 0; per dataset the test bearing with most degradation-phase snapshots). Dotted line: H_c .

TABLE IV

TOP: OUTCOME OF EACH BEARING’S FIRST SUSTAINED (3 SNAPSHOTS) SCHEDULE/URGENT RECOMMENDATION (NUMBER OF BEARINGS, MEAN OVER 3 SEEDS): T TIMELY ($H_c \leq RUL \leq R_{\max}$), L LATE ($RUL < H_c$ OR NEVER), P PREMATURE ($RUL > R_{\max}$); LEAD: MEDIAN RUL AT TIMELY ALARMS (MIN). BOTTOM: SENSITIVITY OF UNSAFE/FM (%) FOR HERA-FULL AND THE POINT POLICY TO THE HORIZONS AND TO $R_{\max}$.

Policy	FEMTO (17 bearings)				XJTU-SY (15 bearings)			
	T	L	P	lead	T	L	P	lead
Always-edge	1.0	0.0	16.0	38	2.0	0.0	13.0	44
Always-cloud	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Point + thr.	1.0	5.3	10.7	40	5.0	4.0	6.0	54
HERA-full	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Width gate	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Setting	HERA		Point		HERA		Point	
$H_c, H_p=0.1, 0.4R_{\max}$	13/68		54/15		16/15		50/5	
$H_c, H_p=0.2, 0.6R_{\max}$	1/95		49/22		9/53		34/8	
$H_c, H_p=0.3, 0.8R_{\max}$	0/100		37/39		2/81		23/42	
$R_{\max} \times \frac{1}{2}$ (seed 0)	24/24		31/23		6/21		54/5	
$R_{\max} \times 2$ (seed 0)	0/100		19/39		0/77		21/31	

ibrates on all non-test bearings, reached 0.94 (edge) and 0.96 (cloud) on FEMTO and 0.93–0.99 on XJTU-SY, at the price of wider intervals (MPIW 2193 s and 3030 s). Per-bearing coverage still varied strongly (lowest cloud coverage 0.30 and 0.07), so exchangeability between bearings remains the binding assumption. Fig. 2 shows intervals narrowing once degradation is visible, far earlier on XJTU-SY.

RQ4: unsafe decisions. Point estimates with fixed thresholds left 49.5% (FEMTO) and 34.2% (XJTU-SY) of critical snapshots without a maintenance action, versus 0.6% and 9.2% for HERA (Table III); the differences were -48.8 pp, 95% CI $[-65.4, -32.0]$ (FEMTO) and -25.0 pp, 95% CI $[-41.8, -10.7]$ (XJTU-SY) over bearings (Wilcoxon $p < 0.001$, $p = 0.004$), and false maintenance rose by 73.3 pp, 95% CI $[60.2, 84.7]$ and 44.9 pp, 95% CI $[17.0, 60.8]$. Sweeping α or a point margin δ (Fig. 3a,c) shows that calibrated and margin-adjusted point policies lie close to one frontier: calibration contributes a principled, held-out choice

of operating point rather than better attainable decisions.

RQ3 and RQ5: escalation and system trade-off. On FEMTO, 90% edge intervals (2193 s on a 3000 s range) almost always straddled H_p , so HERA escalated 97.1% of snapshots, sent more payload than always-cloud streaming (123.0 vs. 112.0 B) and gave no decision benefit. On XJTU-SY, HERA escalated 79.1%, matched always-cloud safety (9.2% vs. 9.2%) with lower false maintenance (53.2% vs. 69.1%), better RMSE (1381 vs. 1421 s), 104.5 instead of 112.0 B/snapshot and mean latency 40.1 vs. 50.6 ms. Two caveats temper this: the FM difference to always-cloud (-15.9 pp, 95% CI $[-35.2, 0.0]$) comes from the two long-lived condition-3 bearings and its interval reaches zero, and the edge-only calibrated policy traced a better frontier than HERA (Fig. 3c) because the conservative edge intervals rarely under-predict. At matched budget, the width gate gave identical results (9.2%/53.3%) and the linear-score gate nearly so (Fig. 3b,d); the anomaly trigger escalated only 5.4% (anomaly AUROC 0.81/0.80). Relaxing α_e to 0.3 reduced escalation to 83.0%/67.9% while unsafe decisions rose to 7.7%/9.2%. Removing the DT context lowered FM on XJTU-SY (25.5%) at higher unsafe rate (12.1%), so the DT context is not uniformly beneficial.

Per-bearing timing and sensitivity. Table IV exposes the main practical limitation: because the 90% lower bound falls below H_p early, every calibrated policy raised its first sustained maintenance alarm while the true RUL still exceeded $R_{\max}$ for 16.0 of 17 FEMTO and 13.0 of 15 XJTU-SY bearings, whereas the point policy was late for 5.3 and 4.0 bearings. Neither is acceptable for scheduling; per-snapshot marginal coverage is the wrong target for alarm timing. In all five horizon settings and all four halved/doubled- $R_{\max}$ runs, HERA left fewer critical snapshots without action than the point policy; with a halved cap on XJTU-SY, HERA reached low false maintenance at the always-cloud unsafe rate (Table IV).

Bearing-level risk control. We also calibrated the alarm offset directly for timing: an onset-gated two-stage alarm whose offset λ was chosen by conformal risk control [14] on the CV+ out-of-fold predictions of the non-test bearings, so that $\Pr(\text{late or missing alarm}) \leq \varepsilon$ per bearing. At $\varepsilon=0.1$ the held-out late rate stayed within the target (2.0% FEMTO, 4.4% XJTU-SY vs. 35.3% and 26.7% for the point policy), but only by alarming early: 94.1% and 80.0% of alarms were premature and the onset gate escalated 99.8% and 98.7% of snapshots. The reason is a heavy tail of required offsets: the median bearing needed an offset of 0.24 (FEMTO) and 0.20 (XJTU-SY) of $R_{\max}$ to be warned in time, but the two least predictable bearings needed 0.67–0.75 and 0.58–0.74. With n calibration units, CRC with a binary loss is feasible only if the fraction π of warning-free units satisfies $\pi < \varepsilon$ and $n \geq (1 - \varepsilon)/(\varepsilon - \pi)$; e.g., $\varepsilon=0.1$ with $\pi=0.05$ needs $n \geq 18$, beyond the ≈ 12 bearings per fold available here.

Detection and compute. Table V shows why no simple trigger suffices: the edge anomaly score reached AUROC 0.81 and 0.80 but recall of only 0.26 and 0.34 at its training-calibrated threshold, whereas the calibrated lower bounds

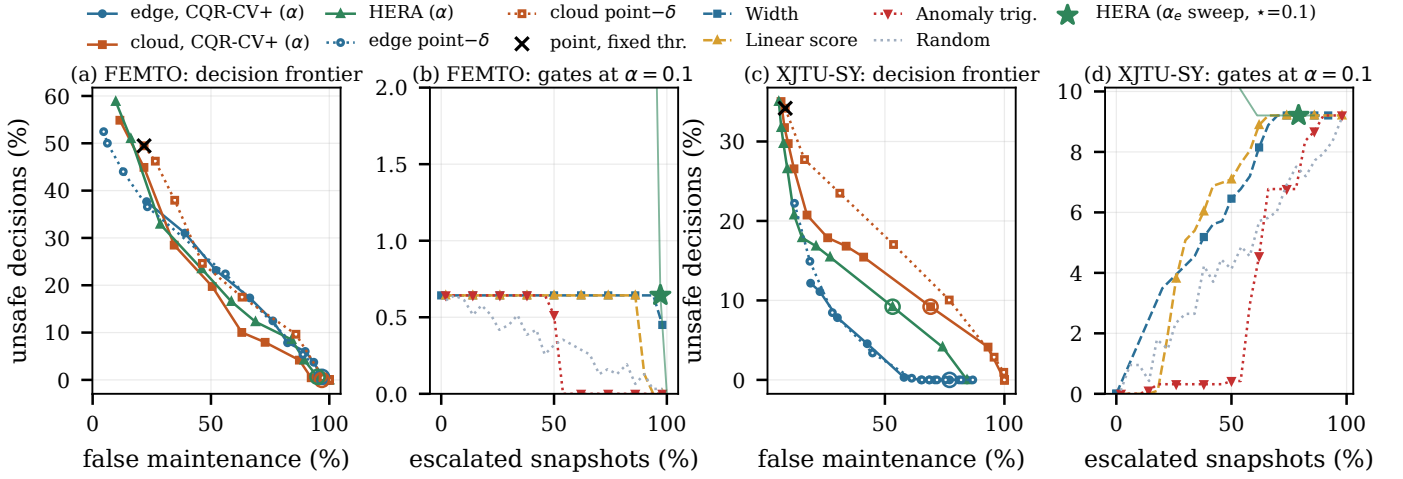


Fig. 3. (a,c) Unsafe decisions vs. false maintenance when sweeping α (calibrated; circles at $\alpha=0.1$) or a margin δ (point). (b,d) Unsafe decisions vs. escalation budget for gates on the same $\alpha=0.1$ edge intervals; green line: HERA with α_e swept.

TABLE V

TOP: DETECTION OF IMMINENT FAILURE ($RUL \leq H_c$) PER SNAPSHOT, THRESHOLDS SET ON TRAINING DATA (MEAN OVER 3 SEEDS). BOTTOM: COMPUTE OF THE SELECTED EDGE AND FOG/CLOUD MODELS (HOST CPU, 1 THREAD, BATCH 1).

Detector	FEMTO			XJTU-SY		
	Prec.	Rec.	AUROC	Prec.	Rec.	AUROC
Anomaly score a_t	0.27	0.26	0.81	0.21	0.34	0.80
Isolation Forest	0.27	0.15	0.77	0.15	0.13	0.81
Edge interval $L_t^e \leq H_c$	0.07	0.68	0.70	0.06	0.94	0.82
Cloud interval $L_t^c \leq H_c$	0.05	0.82	0.65	0.17	0.76	0.81
Model	Params	MACs	ms	Params	MACs	ms
Edge model	2.5k	0.03M	0.15	2.7k	0.02M	0.07
Cloud model	62.1k	3.61M	1.11	13.6k	0.38M	0.62

recalled 0.68–0.94 of critical snapshots at low precision, the safety/false-alarm trade-off seen in Fig. 3. The edge models need tens of thousands of MACs and well under a millisecond; the fog/cloud models need roughly 20–120 times more MACs.

Explainability and oversight. Expected-gradient SHAP on 300 degradation-phase snapshots ranked low-frequency band energies and the spectral centroid (FEMTO) and the DT fast health-index average and mid- and high-frequency band energies (XJTU-SY) highest; additivity errors were 16% and 3% of the explained deviation. An automatically generated card (FEMTO) reads:

Asset Bearing1_3, snapshot 2078. **RUL** 578 s, 90% interval [0, 2138] s. **Anomaly** $a_t=15.8$ (threshold 17.04, no alarm). **Top SHAP** h_band0_1k (-1489 s), v_band2_4k (-1244 s), $h_centroid$ (+1175 s). **Recommendation** urgent maintenance — requires human approval. **Rationale** lower bound $0s \leq H_c$.

All fields come from the predictors via a fixed template. Approval requests covered 95.5% (FEMTO) and 55.8% (XJTU-SY) of HERA snapshots, an operator load that shows calibrated but wide intervals shift burden to humans.

Lessons for deployment. Table VI condenses the evidence.

(i) *Calibrate before routing:* calibrated intervals removed most

TABLE VI

DEPLOYMENT GUIDE DERIVED FROM THE MEASURED RESULTS (FEMTO / XJTU-SY).

Plant condition	Recommended policy	Evidence
Safety-critical, any fleet	Decide on calibrated lower bound	unsafe 0.6%/9.2% vs. 49.5%/34.2% for point thresholds
Degradation visible early	Uncertainty-gated escalation	XJTU-SY: same unsafe rate as always-cloud, FM 53.2% vs. 69.1% (CI reaches 0)
Abrupt, warning-free failures	Always-edge calibrated	FEMTO: HERA escalated 97.1%, no benefit; edge RMSE 753 s best
Limited bandwidth	Gate on interval width	width gate equal to decision-sufficiency gate on both datasets
Alarm-timing guarantee needed	Conformal risk control with $n \geq (1-\epsilon)/(\epsilon-\pi)$ units	with ≈ 12 units: late 2.0%/4.4% but premature 94.1%/80.0%
Limited operator capacity	Relax α or add onset detection	approval load 95.5%/55.8% at $\alpha=0.1$

unsafe point-threshold decisions on both datasets. (ii) *Hierarchy pays only if degradation is observable early:* escalation helped on XJTU-SY and not on FEMTO. (iii) *Small fleets strain calibration:* three calibration bearings under-covered, cross-fitting on all assets fixed pooled coverage, and per-asset coverage still varied. (iv) *The interval width matters more than the gate:* decision-sufficiency and width gates coincided. (v) *Snapshot guarantees are not alarm guarantees:* timely bearing-level alarms need larger fleets. (vi) *Wide intervals move work to humans:* approval load, not model accuracy, limits human-centric use. Limitations: 32 bearings, approximate exchangeability, a state-based DT, host-CPU latency, analytic communication, one-seed $R_{\max}$ sensitivity, and no deployment or user study.

VI. CONCLUSION

Conformal RUL intervals from cross-fitted calibration removed most unsafe decisions of point-threshold maintenance on two bearing datasets. Using them to gate edge-to-cloud escalation paid off only on gradually degrading bearings, where it matched always-cloud safety with less false maintenance and communication; on abruptly failing bearings it escalated almost everything. The specific decision-sufficiency rule was no better than a width gate, and per-snapshot calibration produced premature bearing-level alarms. Calibrating alarm timing directly with conformal risk control met a bearing-level lateness target but, with about a dozen calibration bearings, only by alarming early. Useful timing guarantees will require larger run-to-failure fleets and features with longer warning lead times.

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